

RAWMER: REFERENCE-ASSISTED SPEECH WATERMARKING VIA RELATIVE MEL-ENERGY RELATIONS

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ABSTRACT

Provenance attribution for neural text-to-speech motivates audio watermarking, yet mel-domain reference-assisted schemes that encode each bit as an absolute perturbation pattern degrade sharply under strong additive noise. We propose RAWMER, which encodes each payload bit as the sign of a relative energy difference between two groups of mel bins, repeated across time blocks, with a reliability-aware selection that uses a clean reference to pick boundary-safe, energy-stable, and balanced bin pairs. The differential score cancels common-mode distortion. On HiFi-GAN at matched clean quality, RAWMER raises bit accuracy under 10/5 dB SNR additive noise from 71.3%/63.3% to 94.1%/85.9% over implemented MelShield, with consistent gains on DiffWave. Experiments on relative controls, reference compression and blind partial-clip verification show the setting sound and practical, and we further benchmark against waveform-domain baselines. The main limitation is strong time–frequency desynchronization.

Index Terms— speech watermarking, Provenance verification, Mel spectrogram, Neural text-to-speech, Robust Watermarking

1. INTRODUCTION

Modern text-to-speech (TTS) and voice cloning systems produce speech that is hard to distinguish from genuine recordings, raising risks of spoofing, impersonation and misinformation, and motivating mechanisms that trace audio back to its source [1]. Audio watermarking, which embeds an imperceptible payload that survives later processing, is a central tool for such provenance attribution [2, 3].

Most TTS systems map text to a mel spectrogram and then to a waveform via a vocoder such as HiFi-GAN [4] or DiffWave [5]. Because the mel spectrogram is a shared, model-agnostic interface across these pipelines, recent work moves keyed, in-generation watermarking from the waveform into this domain. MelShield [6] embeds a keyed payload into the intermediate mel spectrogram via low-energy spread-spectrum perturbations before vocoding, so that a

provider retaining the clean reference mel can verify ownership without retraining the vocoder; in our re-implementation it achieves 99.98% bit accuracy under MP3/AAC and 96.85% under 10 dB additive noise at PESQ 3.51. We adopt the same reference-assisted setting: the provider stores a compact clean mel reference at generation time and, given a suspect clip, checks whether it matches a embedded payload, key, and record.

A point of departure is the encoding mechanism. MelShield [6] and prior mel-domain schemes represent each bit as an absolute perturbation pattern, detected by correlating the reference residual with the keyed pattern. Such absolute residuals are sensitive to gain, frequency shaping, and vocoder reconstruction error; at lower SNRs the embedded pattern is buried and bit accuracy falls toward chance. This inability under strong additive noise—not under common compression or filtering—is the problem we target.

Our central observation is that, within a short time block, the relative energy relation between two groups of mel bins is more stable than the absolute residual of any single bin: gain changes, mild filtering, and local reconstruction bias affect a local mel region similarly and largely cancel in a differential measurement. We propose RAWMER, which encodes each payload bit as the sign of an energy difference between a positive and a negative group of mel bins, repeated across time blocks, with a reliability-aware selection strategy that uses the clean reference to choose boundary-safe, energy-stable, and balanced pairs.

We make three contributions:

- **Relative mel-energy watermarking.** We encode payload bits as block-wise relative energy relations between mel-bin groups rather than absolute perturbations, yielding a differential detection score robust to common-mode distortion.
- **Reliability-aware pair selection.** The clean reference is used to pick boundary-safe, energy-stable, and balanced pairs from multiple candidates—the main source of robustness under strong additive noise in our ablations.
- **Systematic study of the setting.** On HiFi-GAN [4] and DiffWave [5], RAWMER substantially improves additive-noise robustness over a quality-matched MelShield [6]

re-implementation; negative controls, references compressed to about 22 KB per utterance, and blind partial-clip verification show the setting is sound and practical. We further benchmark against waveform-domain baselines and identify strong time–frequency desynchronization as the main remaining limitation.

2. METHOD

RAWMER eds an N -bit payload into the intermediate mel spectrogram before vocoding and recovers it from a suspect waveform using a stored clean mel reference.

2.1. Setup

Let $X \in [0, 1]^{F \times T}$ be the per-utterance min–max normalized log-mel and X_B its watermark band of B bins. We partition the time axis into blocks of length L with stride S . Each block k writes a keyed subset $\mathcal{A}_k$ of N_b payload bits, drawn from a permutation seeded by (κ, k) so that every bit recurs across multiple blocks.

2.2. Block-wise Relative Mel-Energy Encoding

For each bit $i \in \mathcal{A}_k$, a ed SHA-256 draw on $(\kappa, \text{uid}, i, k)$ selects $2P$ distinct bins in the band, split into a positive group $\mathcal{P}_{i,k}^+$ and a negative group $\mathcal{P}_{i,k}^-$ of size P . The pair vector $p_{i,k} \in \mathbb{R}^B$ takes value $+1/\sqrt{2P}$ on $\mathcal{P}_{i,k}^+$, $-1/\sqrt{2P}$ on $\mathcal{P}_{i,k}^-$, and zero elsewhere. With bit polarity $s_i = 2m_i - 1$, a block-energy weight $w_k = \mu + (1 - \mu)\tilde{X}_{B,k}^\gamma$ that emphasizes voiced frames, and an active-bit scale $\eta = 1/\sqrt{N_b}$, we accumulate per-block updates into a layer $\ell \in \mathbb{R}^{B \times T}$ with $\ell[\cdot, t] = \sum_{k: t \in [t_k, t_k + L]} \sum_{i \in \mathcal{A}_k} s_i w_k \eta p_{i,k}$. A headroom mask $M = \text{clip}(\min(X_B, 1 - X_B)/\beta, 0, 1)$ suppresses updates near the dynamic-range boundary, and a reuse count $R[b, t] = \sum_{i,k} |p_{i,k}[b]| \mathbb{1}_{[t_k, t_k + L]}(t)$ spreads energy on bins revisited across (i, k) . The watermarked mel is

$$\tilde{X}_B = \text{clip}(X_B + \alpha M \odot \ell / \sqrt{\max(R, 1)}, 0, 1), \quad (1)$$

where α controls embedding strength. The provider stores X_B (or a quantized subset; see §3) as the reference, then vocodes $\tilde{X}$ into waveform.

2.3. Reliability-Aware Pair Selection

A purely random $p_{i,k}$ may place either group on low-energy or near-boundary bins, where the perturbation is absorbed by M or by vocoder reconstruction. For each (i, k) we instead draw C candidate pairs from independent keyed seeds and pick the one whose two groups are both reliable. Let $r[b] = \text{mean}_t [M[b, t](\mu + (1 - \mu)X_B[b, t]^\gamma)]$ over frames in the block,

and let $\rho_j^\pm$ denote the mean of r over the positive/negative group of the j -th candidate. We select

$$j^* = \arg \max_j \min(\rho_j^+, \rho_j^-) - \lambda |\rho_j^+ - \rho_j^-|, \quad (2)$$

with $\lambda = 0.25$: both groups must be reliable *and* balanced in reliability, avoiding pairs whose differential score would be dominated by one side.  single mechanism accounts for most of the additive-noise robustness reported in §3.

2.4. Reference-Assisted Detection

Given a suspect waveform, we recompute the normalized log-mel using the reference statistics and align it to the reference along time by a bounded-shift search ($\pm \Delta_{\max}$ frames) that maximizes the Pearson correlation between frame-mean energy curves, compensating small drift from resampling or compression. After DC-removing the residual $E = \tilde{X}'_B - X_B$, we re-derive the keyed pairs from (κ, uid) and accumulate, for each bit i voted on by V_i blocks,

$$z_i = \frac{1}{\sqrt{V_i}} \sum_{k: i \in \mathcal{A}_k} w_k \eta \langle E_k, p_{i,k} \rangle, \quad (3)$$

where $\langle E_k, p_{i,k} \rangle$ is the joint band–time mean of $E[:, t_k : t_k + L] \odot p_{i,k}$. We decode $\hat{m}_i = \mathbb{1}[z_i \geq 0]$ and verify when $\text{ACC}(\hat{m}, m) \geq \tau$ ($\tau = 0.75$). Since z_i is a differential score over two groups within the *same* local block, common-mode distortions—gain shifts, mild filtering, locally shared reconstruction bias—largely cancel, while balanced pairs render zero-mean noise approximately mean-zero in $\langle E_k, p_{i,k} \rangle$.

3. EXPERIMENTS

3.1. Setup

evaluate RAWMER on LJSpeech [7] with HiFi-GAN [4] and DiffWave [5] vocoders. The mel frontend uses $F = 80$ mel bins and $n_{\text{fft}} = 1024$, with a 22.05 kHz sample rate and a hop size of 256. RAWMER embeds a 32-bit payload with $B = 40$ band bins ($c_{\min} = 20$, $c_{\max} = 60$), $L = S = 8$ frames per block, $N_b = 6$ per block, $P = 6$ bins per group, $C = 16$ candidate pairs, $\alpha = 0.435$ on HiFi-GAN and $\alpha = 0.355$ on DiffWave, $\Delta_{\max} = 12$, $\tau = 0.75$.

We attack the watermarked waveform with MP3/AAC compression, $\pm 8\%$ amplitude scaling, 16 kHz resampling, 300–8000 Hz bandpass / 3 kHz lowpass filtering, additive Gaussian noise at 20/10/5 dB SNR, and 80 ms echo. Bit accuracy (ACC) is the fraction of payload bits recovered; verification rate (VR) is the fraction of utterances with $\text{ACC} \geq \tau$, where $\tau = 0.75$ for RAWMER (i.e., at least 24 of 32 bits must match). Q and STOI are reported against the unmarked vocoder output.

We compare against (i) a re-implemented MelShield [6] ched to RAWMER on clean PESQ ($\alpha = 0.050$ on HiFi-GAN, $\alpha = 0.061$ on DiffWave, 32-bit payload), and (ii)

Table 1. Bit accuracy (ACC) under additive noise on LJSpeech [7]. Mel-domain methods (RAWMER, MelShield [6]) use 32-bit payloads at matched clean PESQ ≈ 3.51 on random 2000 utterances ([2026]); waveform baselines (AudioSeal [2], WavMark [3]) use 16-bit payloads on [500]. [mel-domain entry per column in bold.

Vocoder	Method	none	noise20	noise10	noise5
HiFi-GAN	MelShield	0.999	0.903	0.713	0.633
	RAWMER	1.000	0.994	0.941	0.859
	AudioSeal	1.000	0.974	0.790	0.647
	WavMark	1.000	0.752	0.511	0.493
DiffWave	MelShield	1.000	0.969	0.823	0.722
	RAWMER	1.000	0.990	0.922	0.835
	AudioSeal	1.000	0.966	0.754	0.618
	WavMark	1.000	0.670	0.512	0.498

[ic AudioSeal [2] and WavMark [3] models as cross-paradigm waveform baselines (16-bit payload, post-vocoder embedding).

3.2. Main Comparison

Table 1 reports robustness at matched clean quality. On HiFi-GAN, RAWMER lifts $\text{ACC}_{\text{noise10}}$ from 71.3% to 94.1% and $\text{ACC}_{\text{noise5}}$ from 63.3% to 85.9% over MelShield; on DiffWave, from 82.3% to 92.2% and 72.2% to 83.5%. Other non-noise attacks (MP3, AAC, scaling, 16 kHz resampling, bandpass, lowpass, echo) remain at $\text{ACC} \geq 0.983$ for all methods and are [itted for space.

AudioSeal and WavMark have [er clean PESQ (4.0–4.5 vs ~ 3.51 for mel-domain methods) but qualitatively different additive-noise behavior. AudioSeal preserves moderate bit accuracy yet its detector rejects most noisy samples (VR drops to 0.14–0.30 at 20 dB and to zero at 10/5 dB). WavMark’s high-level decoder declines to emit a payload on $\geq 95\%$ of noisy samples; even with forced decoding, ACC collapses to ~ 0.50 at 10/5 dB. These reflect different deployment trade-offs and payload regimes (16 vs 32 bits), but in our reference-assisted, noise-heavy setting, RAWMER yields the [est usable verification.

[ification under matched threshold. RAWMER’s threshold $\tau = 0.75$ (24 of 32 bits) [rictor than MelShield’s reported $\tau = 0.61$, so a natural concern about Table 1 is whether this margin alone explains the gap. Table 2 relaxes this by reporting VR under both thresholds on both vocoders (ACC is independent of τ). Two observations stand out. First, MelShield’s reported VR depends heavily on τ : on HiFi-GAN at $\tau = 0.75$, its VR collapses to 0.434 at 10 dB and 0.127 at 5 dB, well below RAWMER’s 0.991 and 0.918 at the same threshold. Second, RAWMER at the relaxed $\tau = 0.61$ [hes near-perfect VR (0.985–0.995 at 5 dB on both vocoders), showing that its bit-level margin translates

Table 2. Verification rate (VR) under two thresholds on LJSpeech (random 2000). ACC is independent of τ and reported in Table 1. Best entry per (vocoder, τ , attack) cell in bold.

Vocoder	τ	Method	noise20	noise10	noise5
HiFi-GAN	0.75	MelShield	0.969	0.434	0.127
		RAWMER	1.000	0.991	0.918
	0.61	MelShield	0.999	0.874	0.612
		RAWMER	1.000	1.000	0.995
DiffWave	0.75	MelShield	0.997	0.841	0.485
		RAWMER	1.000	0.983	0.871
	0.61	MelShield	1.000	0.973	0.872
		RAWMER	1.000	1.000	0.985

into usable verification across a range of operating thresholds.

3.3. Reference-Assisted Verification

Negative controls. [probe whether the reference-assisted setting enables trivial false positives on 2000 unattacked HiFi-GAN utterances. With the correct reference, key, and claimed payload, RAWMER passes at $\text{VR} = 1.000$. Substituting a wrong key, a wrong claimed payload, or a randomly selected reference returns $\text{ACC} \in [0.499, 0.503]$ and $\text{VR} \in [0\%, 0.40\%]$. Notably, wrong-payload trials retain detector confidence on par with positives but their decoded bits disagree with the claim, confirming that verification must hinge on payload match, not confidence magnitude.

Reference compression. The stored reference is the band X_B , not the full mel. Table 3 shows that quantizing X_B to 8 bits per sample shrinks the reference to about 22 KB—about 1/8 of float32—with [ntially no robustness loss. Spending the same 22 KB on a coarse 4-bit full-mel reference is strictly worse: precision inside the watermark band dominates coverage outside it. Pushing further to 4-bit band-only quantization (about 11 KB, about 1/16 of float32) still retains 0.913/0.825 ACC at 10/5 dB SNR, confirming that the band-localized differential detector tolerates aggressive compression on the relevant frequency range. In our experiments the same compression strategy is empirically more disruptive to MelShield.

Blind partial-clip verification. Because each payload bit is repeated across keyed blocks, RAWMER survives cropping in time. We test this without revealing the crop offset: a 25% slice from the start, middle, or end of the watermarked audio is slid across the reference (stride 4 frames) and the window with the largest signed score is taken. The selected window’s median start error is 0–1 mel frames, ACC stays in 0.985–0.991, and $\text{VR} \geq 0.998$ for all three positions; 50% clips reach $\text{VR} = 1.000$.

Table 3. Reference  expression on HiFi-GAN [4] RAWMER (random 2000), reported as ACC/VR. “Band-only” stores quantized values for the watermark band only; “full-mel” stores all 80 bins.

Variant	KB	noise20	noise10	noise5
 §2 (full-mel)	175.0	0.994 / 1.000	0.941 / 0.994	0.859 / 0.931
 (band-only)	21.9	0.994 / 1.000	0.939 / 0.993	0.858 / 0.925
uint4 (full-mel)	21.9	0.982 / 1.000	0.905 / 0.975	0.818 / 0.842
uint4 (band-only)	11.0	0.983 / 1.000	0.913 / 0.984	0.825 / 0.856

Table 4. Candidate count C on HiFi-GAN [4] RAWMER (random 2000). Noise columns are ACC.

C	PESQ _{clean}	noise20	noise10	noise5
1	3.55	0.946	0.767	0.668
4	3.53	0.987	0.889	0.785
8	3.52	0.992	0.923	0.832
16	3.51	0.994	0.941	0.859

3.4. Ablation and Robustness Boundary

Reliability-aware pair selection. Table 4 sweeps the candidate count C from which the most reliable balanced pair is chosen. Going from $C = 1$ (a single random pair) to $C = 16$ lifts ACC_{noise5} from  % to 86.1% with negligible PESQ cost. This is the single largest contributor to the noise robustness reported above and isolates the value of the selection step in Eq. (2).

Robustness boundary.  extended attacks. RAWMER remains robust to 64 kbps MP3, 48 kbps AAC, 2 kHz low-pass, 8-bit quantization, ± 0.5 amplitude clipping, 100 Hz–7 kHz bandpass, and stronger reverb (ACC ≥ 0.987 on both vocoders), as well as  codec [?] re-encoding at 6/12/24 kbps (ACC ≥ 0.976). The genuine limit is time-frequency desynchronization: pitch shifts within ± 50 cents stay above 99% ACC, but ± 100 cents drops to 78%; speed changes of $\pm 3\%$ retain VR ≥ 0.996 , $\pm 5\%$ degrades to 0.85–0.91, and $\pm 10\%$  below 0.32. These failures arise from the bounded-shift alignment in §2.4 breaking down, not from the watermark energy being lost—multi-scale or DTW-based alignment is a natural extension.

4. CONCLUSION

We presented RAWMER, a reference-assisted mel-domain speech watermark that encodes payload bits as block-wise relative energy relations between reliability-selected mel-bin groups. On LJSpeech with HiFi-GAN and DiffWave, at matched clean PESQ, RAWMER substantially outperforms a re-implemented  Shield under strong additive noise while maintaining low false positives under wrong key, payload, and reference, supporting compressed references of

about 22 KB per utterance and blind partial-clip verification. The main remaining limitation is robustness to large speed changes and pitch shifts, which break the frame-level reference alignment; multi-scale or DTW-based matching is a natural extension we leave to future work.

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